

Ayudantia Control

IEE3311

PUC

2026

Ejercicio

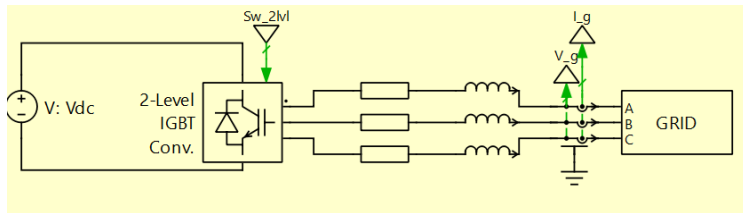


Fig. 1: Conversor conectado a la red

Potencia con circuito desbalanceado

La potencia aparente considerando un sistema desbalanceado considerando secuencia positiva y negativa es:

$$P_g + jQ_g = k_{\alpha\beta}(\vec{v}_{g\alpha\beta}^p + \vec{v}_{g\alpha\beta}^n)(\vec{i}_{g\alpha\beta}^p + \vec{i}_{g\alpha\beta}^n)^c \quad (1)$$

Expandiendo queda:

$$P_g(t) = P_{g0} + P_{gc2} \cos(2\omega_g t) + P_{gs2} \sin(2\omega_g t) \quad (2)$$

$$Q_g(t) = Q_{g0} + Q_{gc2} \cos(2\omega_g t) + Q_{gs2} \sin(2\omega_g t) \quad (3)$$

Potencia con circuito desbalanceado

$$P_{g0} = k_{\alpha\beta}(v_{g\alpha}^p i_{g\alpha}^p + v_{g\beta}^p i_{g\beta}^p + v_{g\alpha}^n i_{g\alpha}^n + v_{g\beta}^n i_{g\beta}^n) \quad (4)$$

$$P_{gc2} = k_{\alpha\beta}(v_{g\alpha}^p i_{g\alpha}^n + v_{g\beta}^p i_{g\beta}^n + v_{g\alpha}^n i_{g\alpha}^p + v_{g\beta}^n i_{g\beta}^p) \quad (5)$$

$$P_{gs2} = k_{\alpha\beta}(v_{g\beta}^n i_{g\alpha}^p - v_{g\alpha}^n i_{g\beta}^p - v_{g\beta}^p i_{g\alpha}^n + v_{g\alpha}^p i_{g\beta}^n) \quad (6)$$

$$Q_{g0} = k_{\alpha\beta}(v_{g\beta}^p i_{g\alpha}^p - v_{g\alpha}^p i_{g\beta}^p + v_{g\beta}^n i_{g\alpha}^n - v_{g\alpha}^n i_{g\beta}^n) \quad (7)$$

$$Q_{gc2} = k_{\alpha\beta}(v_{g\beta}^p i_{g\alpha}^n - v_{g\alpha}^p i_{g\beta}^n + v_{g\beta}^n i_{g\alpha}^p - v_{g\alpha}^n i_{g\beta}^p) \quad (8)$$

$$Q_{gs2} = k_{\alpha\beta}(v_{g\alpha}^p i_{g\alpha}^n + v_{g\beta}^p i_{g\beta}^n - v_{g\alpha}^n i_{g\alpha}^p - v_{g\beta}^n i_{g\beta}^p) \quad (9)$$

Potencia en forma matricial

$$\begin{bmatrix} P_{g0}^* \\ Q_{g0}^* \\ P_{gs2} = 0 \\ P_{gc2} = 0 \end{bmatrix} = k_{\alpha\beta} \begin{bmatrix} v_{g\alpha}^p & v_{g\beta}^p & v_{g\alpha}^n & v_{g\beta}^n \\ v_{g\beta}^p & -v_{g\alpha}^p & v_{g\beta}^n & -v_{g\alpha}^n \\ v_{g\beta}^n & -v_{g\alpha}^n & -v_{g\beta}^p & v_{g\alpha}^p \\ v_{g\alpha}^n & v_{g\beta}^n & v_{g\alpha}^p & v_{g\beta}^p \end{bmatrix} \begin{bmatrix} i_{g\alpha}^p \\ i_{g\beta}^p \\ i_{g\alpha}^n \\ i_{g\beta}^n \end{bmatrix} \quad (10)$$

Corrientes de referencia

$$i_{g\alpha}^{p*} = \frac{1}{k_{\alpha\beta}} \left[\frac{v_{g\alpha}^p P_{g0}^*}{v_{g\alpha}^{p^2} + v_{g\beta}^{p^2} - v_{g\alpha}^{n^2} - v_{g\beta}^{n^2}} + \frac{v_{g\beta}^p Q_{g0}^*}{v_{g\alpha}^{p^2} + v_{g\beta}^{p^2} + v_{g\alpha}^{n^2} + v_{g\beta}^{n^2}} \right] \quad (11)$$

$$i_{g\beta}^{p*} = \frac{1}{k_{\alpha\beta}} \left[\frac{v_{g\beta}^p P_{g0}^*}{v_{g\alpha}^{p^2} + v_{g\beta}^{p^2} - v_{g\alpha}^{n^2} - v_{g\beta}^{n^2}} - \frac{v_{g\alpha}^p Q_{g0}^*}{v_{g\alpha}^{p^2} + v_{g\beta}^{p^2} + v_{g\alpha}^{n^2} + v_{g\beta}^{n^2}} \right] \quad (12)$$

$$i_{g\alpha}^{n*} = \frac{1}{k_{\alpha\beta}} \left[\frac{-v_{g\alpha}^n P_{g0}^*}{v_{g\alpha}^{p^2} + v_{g\beta}^{p^2} - v_{g\alpha}^{n^2} - v_{g\beta}^{n^2}} + \frac{v_{g\beta}^n Q_{g0}^*}{v_{g\alpha}^{p^2} + v_{g\beta}^{p^2} + v_{g\alpha}^{n^2} + v_{g\beta}^{n^2}} \right] \quad (13)$$

$$i_{g\beta}^{n*} = \frac{1}{k_{\alpha\beta}} \left[\frac{-v_{g\beta}^n P_{g0}^*}{v_{g\alpha}^{p^2} + v_{g\beta}^{p^2} - v_{g\alpha}^{n^2} - v_{g\beta}^{n^2}} - \frac{v_{g\alpha}^n Q_{g0}^*}{v_{g\alpha}^{p^2} + v_{g\beta}^{p^2} + v_{g\alpha}^{n^2} + v_{g\beta}^{n^2}} \right] \quad (14)$$